

Virtual Office

A Multi-User Immersive Collaborative Workspace in VR

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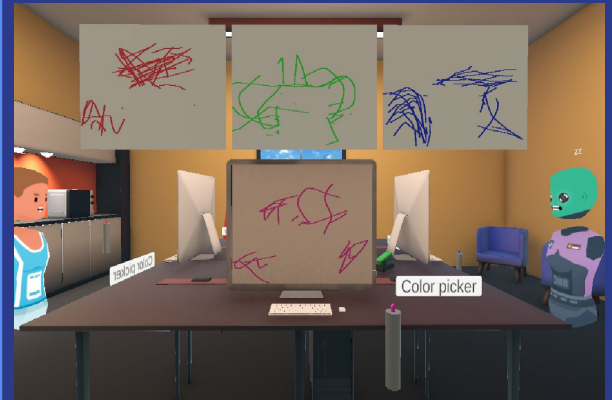
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Unity 6

Ubiq 1.0

Meta Quest 3

C#



The Problem: Sequential Visual Collaboration

Sequential Screen Sharing

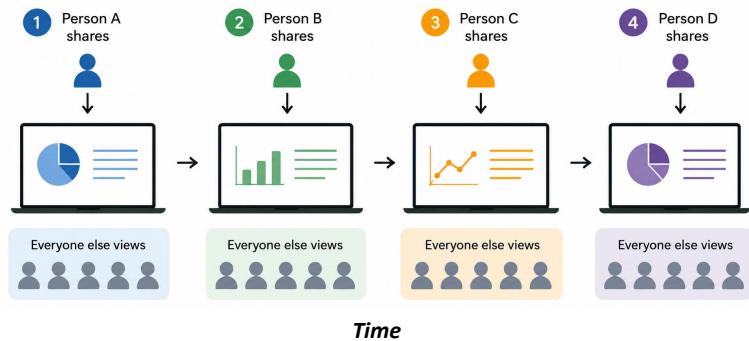
In Zoom, Google Meet and Teams only one user at a time can share their screen.

Constant Handoffs

Collaborative sessions degenerate into a series of presenter handoffs between participants.

Loss of Context

It's impossible to monitor others' work in real time, reducing the perception of a shared space.



Issue: 1 presenter at a time

Virtual Office: Our Answer

A shared virtual office in VR where up to 4 remote users collaborate in real time, each at their own workstation with a personal whiteboard and virtual monitors that are always visible.



Personal Workstation

Each user has a dedicated desk, whiteboard and monitors



Digital Whiteboard

Virtual pen with HSV color picker and eraser



Always-On Visibility

Monitors show other users' whiteboards via Render Textures



Spatial Audio

Voice built in with Ubiq, no extra setup required

System Architecture

Hardware

Meta Quest 3 with controllers

XR Interaction

XR Interaction Toolkit 3.0.7

Engine

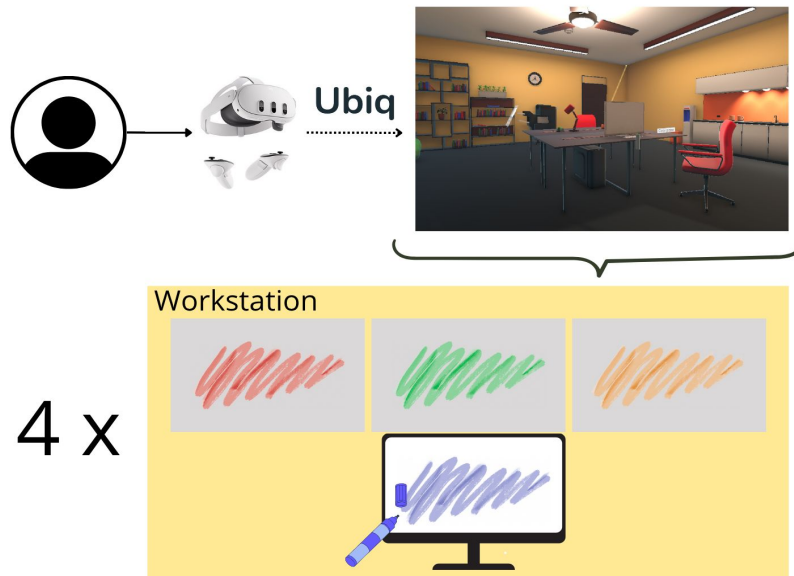
Unity 6.0 · C# Scripts

Networking

Ubiq 1.0.0-pre16 · Local server

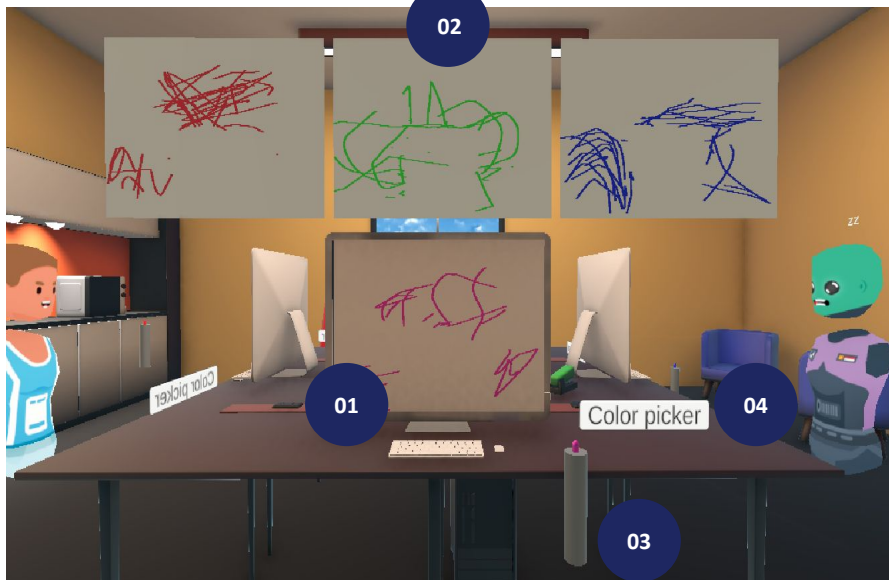
Assets & Tools

Office Pack · Skybox Series · VSCode



System architecture

The Workstation: Core of the Experience



First-person view

01 Whiteboard

2D plane: 1920×1080 resolution, **writable**.
Synchronized over the network.

02 Virtual Monitors

3 planes sampling the other users' whiteboards with zero additional overhead.

03 WhiteboardMarker

Interactable Pen. Allow to write from the nib.

04 Color Picker

Button that opens a floating HSV panel to change the color.

Networking & Synchronization Protocols



Late-Join Snapshot

- › The peer with the smallest UUID (master) is elected deterministically.
- › The texture is JPEG-compressed, base64-encoded, and split into 24 KB chunks.
- › The new peer reassembles the chunks and immediately shows the current whiteboard.



Slot Assignment

- › Each peer reads the others' Ubiq properties after a short delay (coroutine).
- › It picks the first free slot (1-4) and writes it to its own peer property.
- › Avoids the race condition where everyone ended up in slot 1.



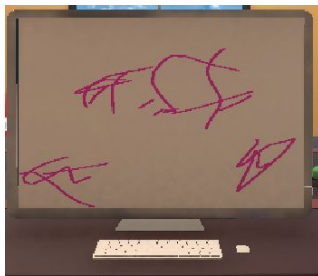
Stroke Interpolation

- › The network message includes the current position and the last point.
- › The receiving peer linearly interpolates between the two points.
- › Produces continuous strokes regardless of the network tick rate.

Custom C# Scripts

`{}` **Whiteboard.cs**

Real-time stroke synchronization + late-join protocol with JPEG → base64 chunking.



`{}` **WhiteboardMarker.cs**

XR grab, raycast UV → pixel mapping, interpolation, owner/non-owner pattern for pen-position sync.



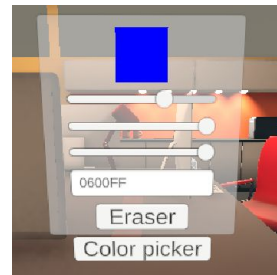
`{}` **WorkstationManager.cs**

Listens to Ubiq events (OnJoinedRoom, OnPeerAdded, OnPeerRemoved) + slot-assignment coroutine + dynamic visibility.



`{}` **ColorPickerUI.cs**

HSV color picker with sliders, hex input, live swatch preview and dynamic binding to the active marker.



Challenges Faced & Solutions Adopted

⚠ Race condition in slot assignment



✓ A coroutine delay before reading peer properties: the new peer waits for already-occupied slots to propagate.

⚠ Discontinuous strokes over the network



✓ Sending last_point with every message + linear interpolation on the receiver → smooth strokes at any tick rate.

⚠ All workstations always visible



✓ WorkstationManager dynamically activates only the workstations matching connected users.

⚠ Color picker with no target pen



✓ Marker ownership system: the ColorPickerUI registers the closest grabbed pen and applies color changes to it.

Evaluation & Results

✓ Functional Tests Passed

- ✓ Single user: whiteboard rendering, pen grab/release, color picker
- ✓ 2 users: real-time stroke sync + Virtual_Whiteboard mirroring
- ✓ 3-4 users: workstation activation, slot assignment, correct visibility
- ✓ Late join: snapshot correctly transmitted and applied

⚠ Known issue: on disconnect, the last workstation is deactivated instead of the one belonging to the leaving peer.

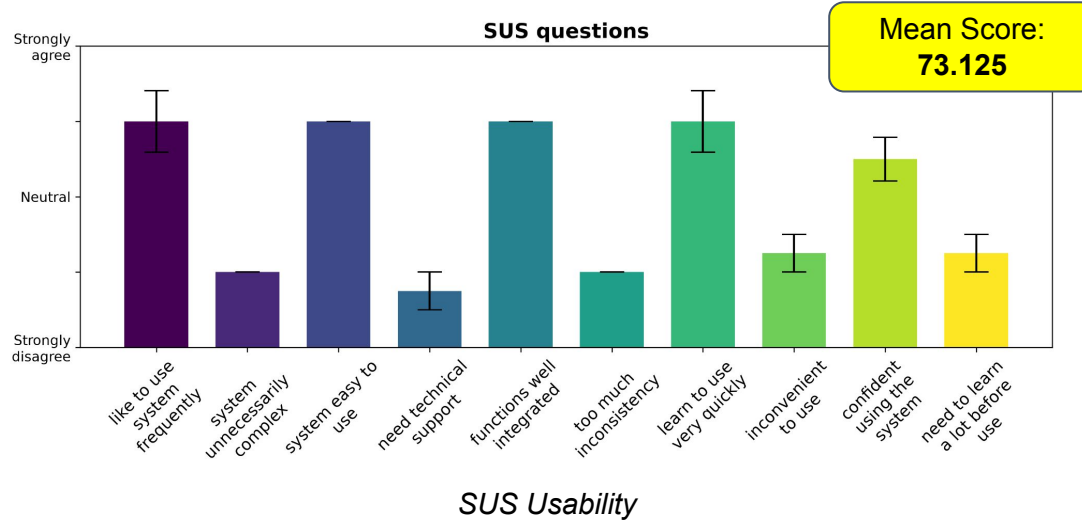
💬 Usability Observations (on PC)

- Experience described as genuinely immersive compared to a standard video call.
- Always-on whiteboard visibility was the most appreciated feature.
- Mouse pen control is awkward in flat-screen mode (expected for desktop/testing use).
- HSV color picker intuitive and easy to use.

Evaluation & Results



Formal Evaluation: SUS Usability & SUS Presence



Participant	Mean score
P1	3.50
P2	3.17
P3	4.23
P4	3.17
Mean	3.42

SUS Presence

Conclusions & Future Work

Key Results

- Main goal achieved: presenter handoffs eliminated.
- Built a collaborative, social, multi-user architecture on Unity 6 + Ubiq with open-source tools.
- Up to 4 simultaneous users: positive formal tests, immersive experience.
- Deliberate simplification (whiteboard vs. desktop stream), maintainable for future versions.

Future Work

Proper disconnect handling: UUID → slot mapping

More formal study with Meta Quest 3 hardware

Support more than 4 users with a dynamic workstation layout

Replace whiteboard with a real desktop video stream (Unity screen capture)

**Thank you for
your attention**